

Certainty Thermometer Teacher Notes

TOK Cognitive Science Activity Bank • Cross-AOK • 25-35 min

Category	Cross-AOK	Time	25-35 min
Difficulty	Easy	ID	certainty-thermometer

Big idea: Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

Overview

Students place claims from different AOKs on a certainty scale, then revise the scale after adding criteria.

Student Prompt

Inspect Certainty Thermometer Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Certainty Thermometer Stimulus A	Student-facing stimulus 1: Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'. Name the AOK, the method being used, and the standard of evidence you would apply.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Certainty Thermometer Stimulus B	Student-facing stimulus 2: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Use this for comparison, a second round, or a partner challenge.
Certainty Thermometer Reveal / Twist	Student-facing stimulus 3: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Teacher reveal: connect the changed response to the big idea — Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Certainty Thermometer Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Certainty Thermometer Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	Is certainty the best measure of knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how certainty is not one simple measure; it depends on evidence, method, context, and purpose.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Make the certainty scale physical if possible, using the wall or floor.
- Start with intuitive placement before introducing criteria cards.

Ready-to-Use Examples

- Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'.
- Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups claim cards and ask them to place each on a certainty thermometer.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Groups explain the criterion they used first.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Introduce criteria cards such as proof, replication, interpretation, consensus, usefulness, and ethical stakes.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students revise placements using at least two criteria.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why certainty differs across knowledge contexts.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Certainty is not the only value knowledge can have.
- Which claim moved when you changed the criterion?

Common Misconceptions

- Students may apply one AOK's standard of evidence to all knowledge claims.
- Students may equate certainty with value or usefulness.

Assessment Look-Fors

- Compares standards of evidence across at least two AOKs.
- Explains how a claim changes when method, context, or audience changes.

Differentiation and Adaptations

- Support: use five claim cards.
- Challenge: require students to defend two different placements for the same claim.

Extension Tasks

- Students build separate thermometers for certainty, usefulness, and ethical importance.
- Use the activity to plan a TOK essay comparison.

Related Activities

- Reliability Ranking
- Claim Migration Lab
- What Counts as Evidence?